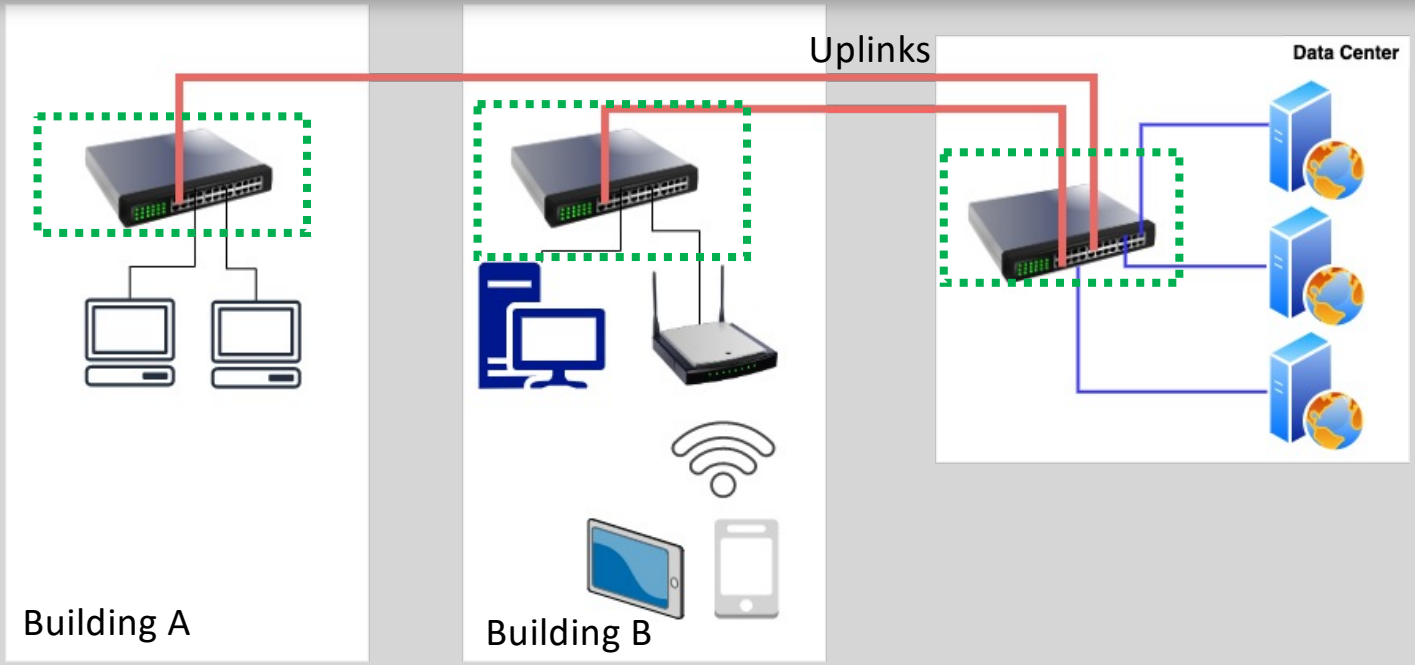
Campus LANs & Switches

Campus LAN

- Switches can be used to build a LAN in a campus (school, university, corporate)
- We interconnect switches using uplinks to form the campus LAN



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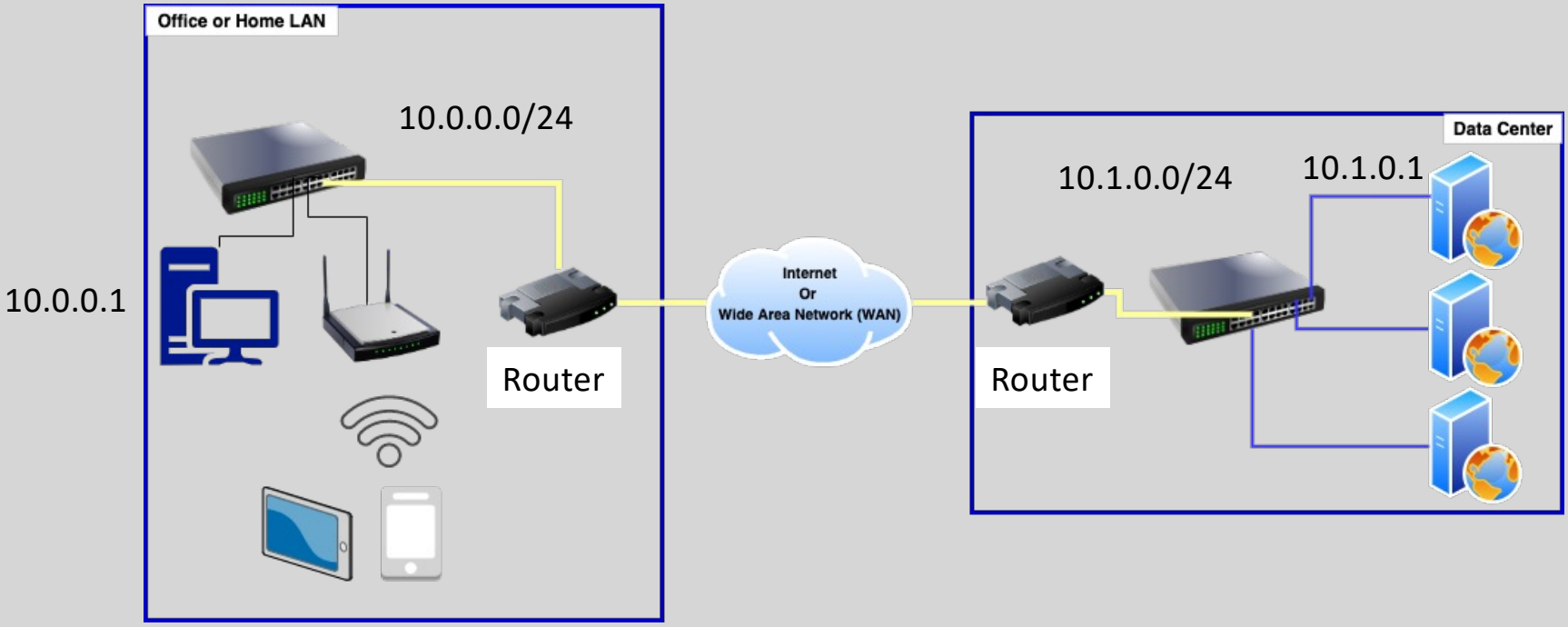
Routers

Routers are required to connect separate IP Networks or Subnets



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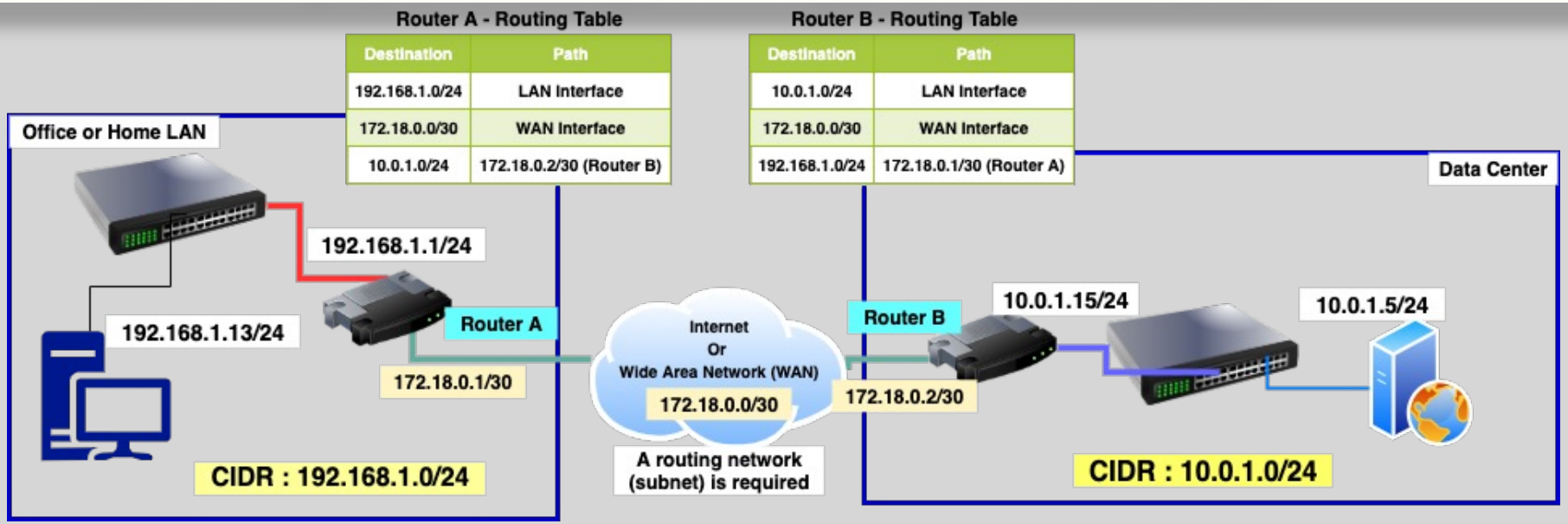
IP Routing – How?

- Each router builds a routing table (database) of known destinations and how to reach them
- Routing tables can be configured statically (manually) – Static IP routing
- Routing tables can also be configured dynamically – Dynamic IP routing
  - ➢ Requires using Dynamic routing protocols (OSPF, BGP are examples)



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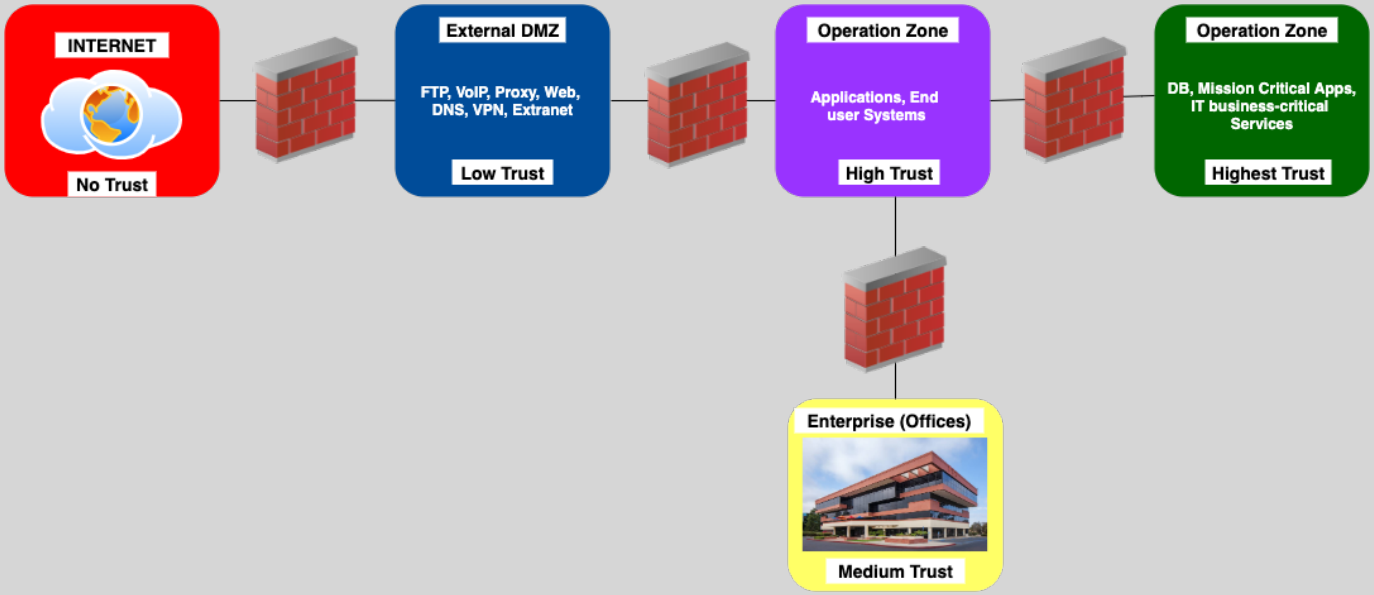
Infrastructure Security – Zoning Architecture

- The security architecture of the network infrastructure consists of zones.
- Zoning architecture depends on how critical and the security risk of the different components.



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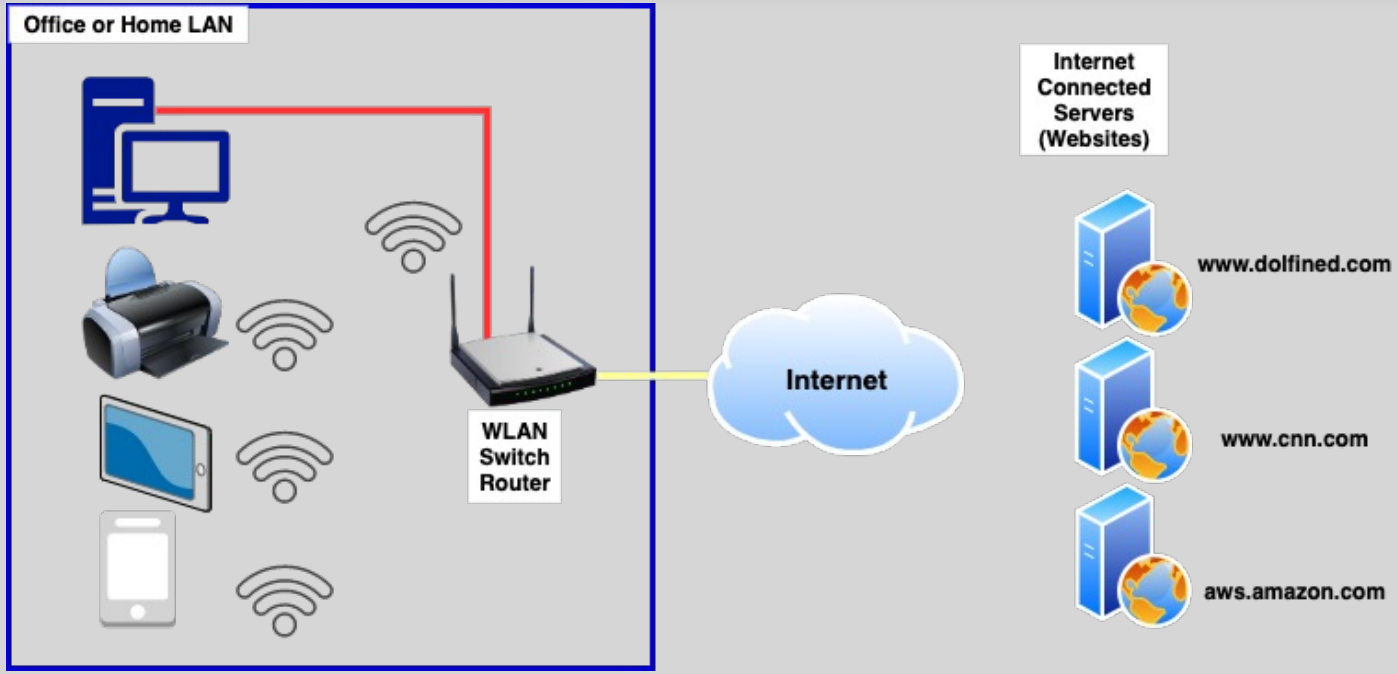
Router/Switch Devices

- Your home WiFi device has WLAN, Switch, and Router functionalities
- When two devices on the same subnet or IP network need to communicate, they do not need to consult the routers (example printing on a WiFi-connected printer at home)



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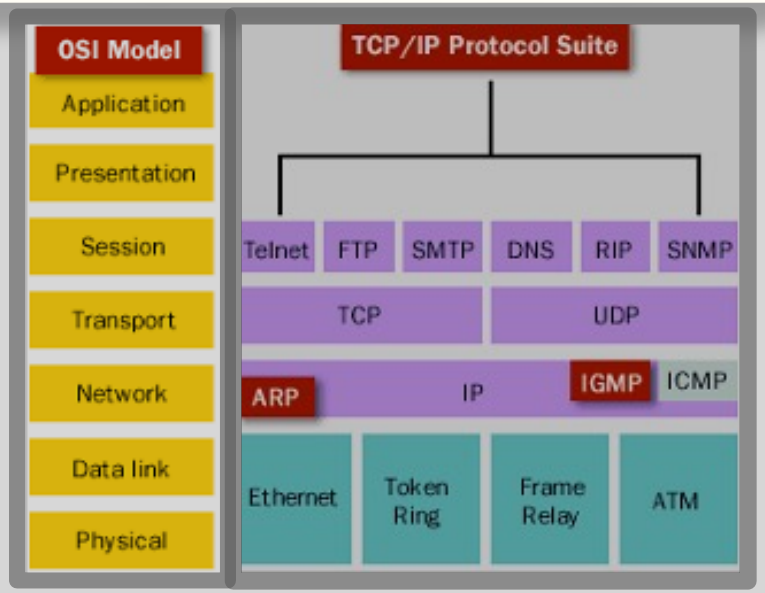
OSI Reference Model vs. TCP/IP Model

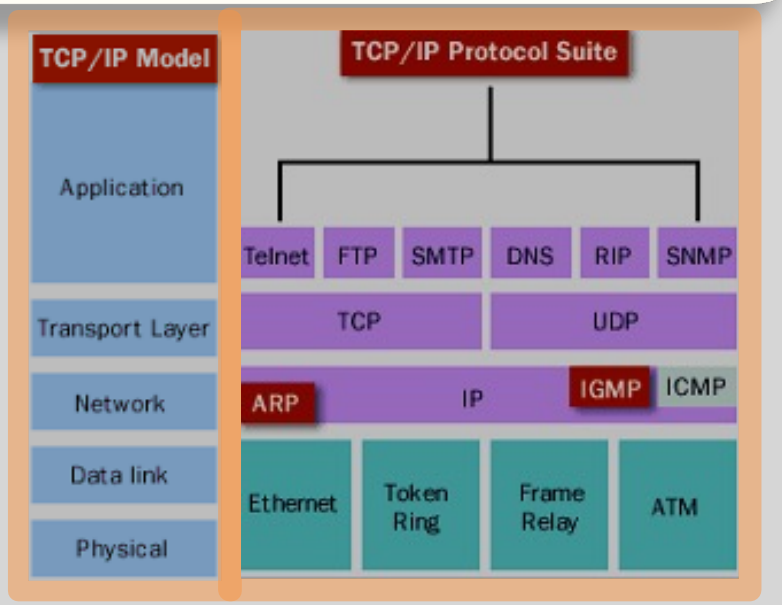
It was designed to allow for interoperability between different vendors in data communications.

- 7 Layers.
- Includes specifications for each layer.

Consists of a four-layer reference model.

- Layers 1-4 correspond to the same layers as OSI Model
- Layer 4 corresponds to layers 5,6,7 of the OSI Model





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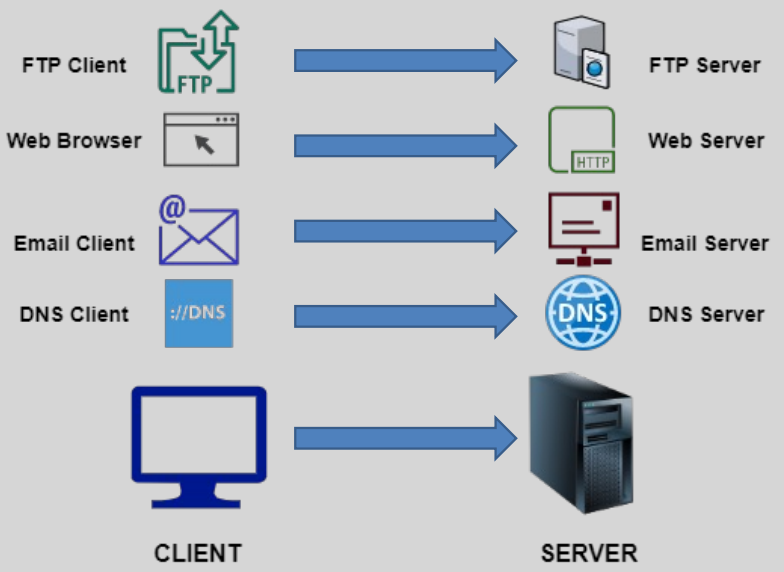
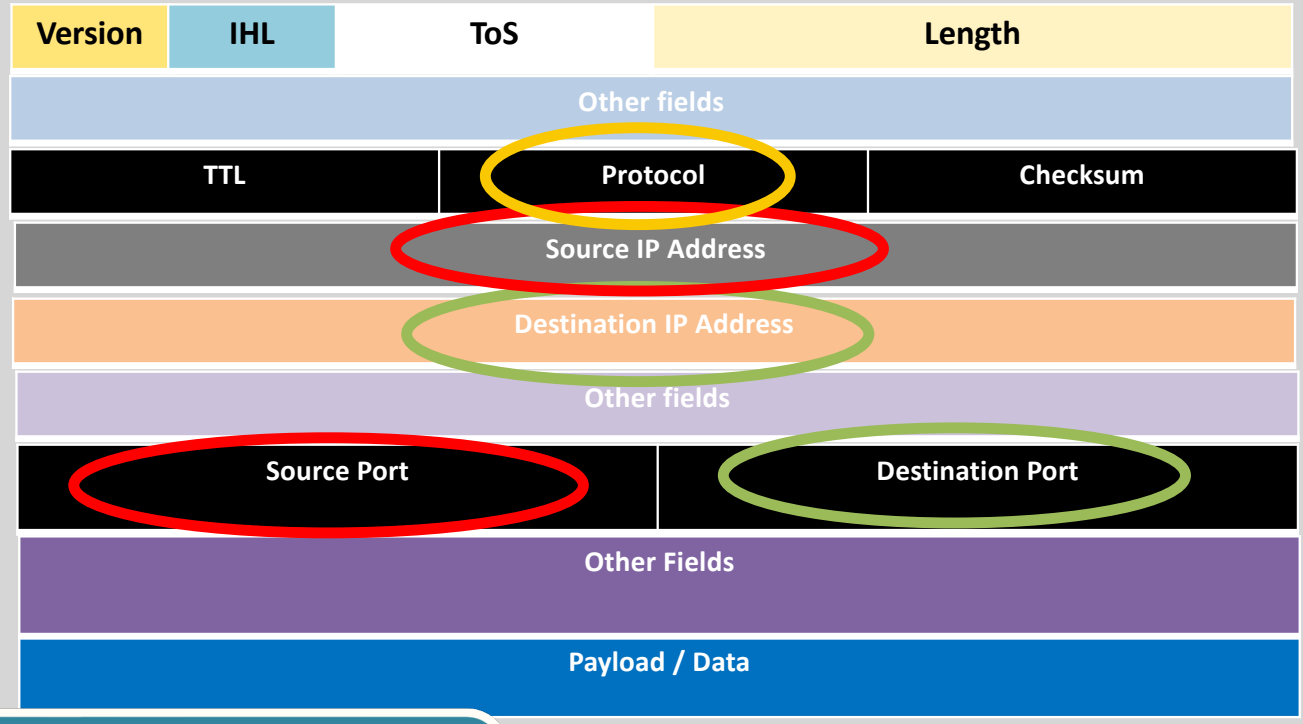
Anatomy of a TCP/IP Packet - Simplified





- Protocol : TCP =6, UDP=17
- Source Address/Port: Sender's IP address/Port
- Destination Address/Port: Receiver's IP address/Port

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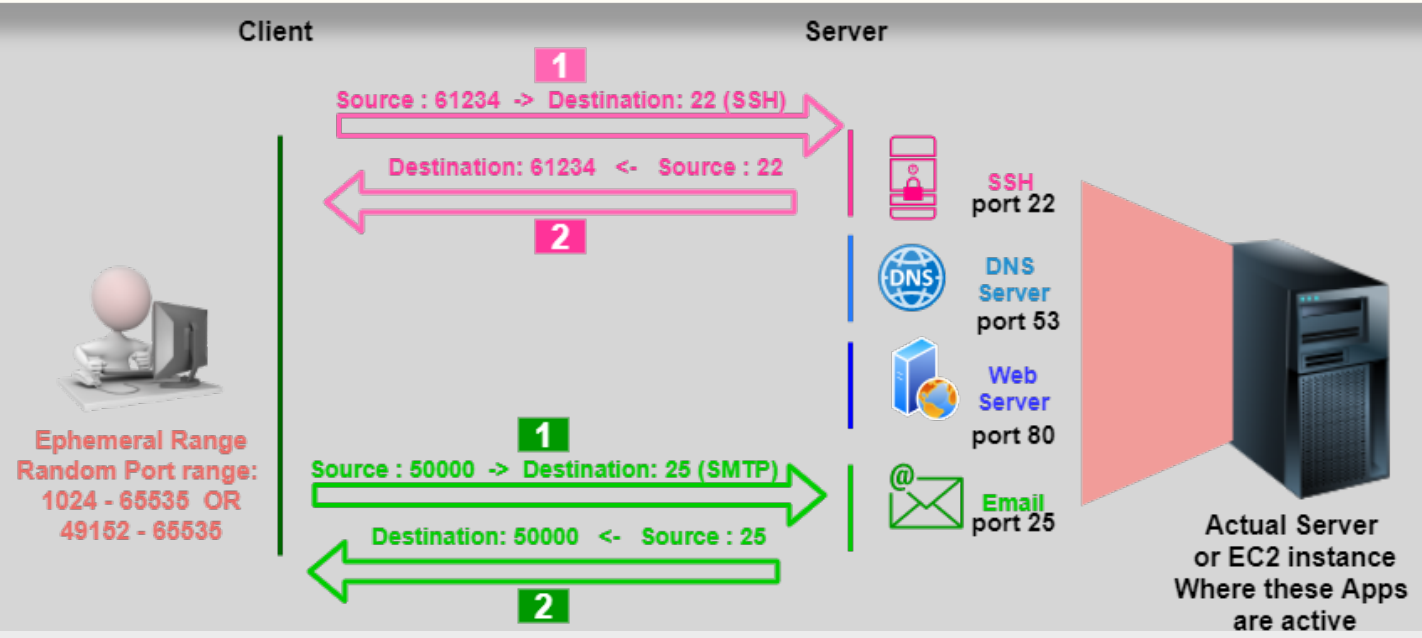
Client to Server Communications – Understanding Source/Destination Ports!

From Client to Server:

- Destination port is the port the application at the destination server is listening on.
- Source port is picked randomly/dynamically from an available range (1024-65535 OR 49152-65535).



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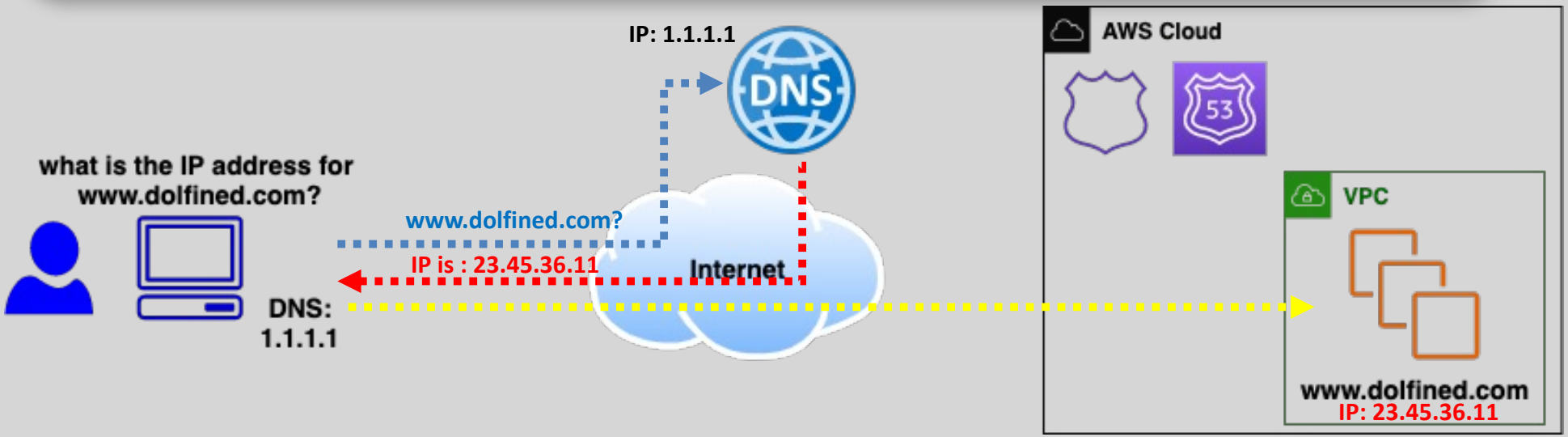
Domain Name Service (DNS) – Why do we need it?

- It is much easier to remember a domain name than an IP address.
- Websites and similar destinations on the internet use IP addresses.
- Domain Name Servers (DNS) are databases on the internet mapping domain names to IP address(es).
- Each device connected to the internet must have a DNS in its TCP/IP configuration.



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